

Safety of Household Electrical Appliances (IS 302 family)

Scope

The IS 302 series specifies safety requirements for household and similar electrical appliances to protect users against electric shock, thermal hazards and mechanical hazards. Part 2-1 states general requirements; the Part 2-x sub-standards add requirements for particular appliance categories such as washing machines (2-11) and heating appliances for liquids (2-15).

Key Requirements

Appliances are assessed for insulation, earthing continuity, touch current and abnormal operation. Heating circuits require over-temperature protection and boil-dry cut-offs. Where relevant, electromagnetic compatibility is verified. Marking must include rated voltage, frequency, power rating, and the manufacturer's identity. Appliances under compulsory certification must carry the Standard Mark (ISI mark) only through licensed manufacturers.

Compliance & Testing

Type tests are performed on samples drawn by the certification body. Surveillance testing of marked product is carried out from the market. Compliance with the series is demonstrated by successful completion of the prescribed test schedule in a recognised laboratory. The scope of each Part 2-x standard lists the appliance categories and exclusions.